

Roll No.....

MID SEMESTER EXAMINATION MARCH- 2025

Name of the Program: B.Tech.

Semester: Eighth

Name of the course: Tribology Paper Code: **TME 815**

Time: 90 Minutes

Maximum Marks: 50

Note: -

- (i) Answer all questions by choosing any one of the subsections
(ii) Each question carries 10 marks

Q1.	10 marks	
(a)	What is Tribology and explain its significant towards sustainability with suitable examples. Also, list various application of Tribology.	CO1
(b)	Explain Newtonian and non-Newtonian fluids with appropriate graphs and examples.	
Q2.	10 marks	
(a)	What is Tribological system and explain its properties in detail.	C01
(b)	A lubricating oil has a kinematic viscosity of 90 cSt (Centistokes) at 40°C and 15 cSt at 100°C. Using the ASTM method, determine the Viscosity Index (VI) of the oil, given that the values of L and H from standard tables are 110 cSt and 12 cSt, respectively.	
Q3.	10 marks	
(a)	Explain the following in detail: 1. Ostwald Viscometer 2. Rotational Viscometer. 3. Falling Sphere Viscometer 4. Brookfield Viscometer	C01/ CO2
(b)	Describe the mechanisms of adhesion and deformation in friction theory	
Q4.	10 marks	
(a)	Explain the following theories of friction and their limitations in detail: Mechanical Interlocking, Molecular attraction, Electrostatic forces, welding shearing and ploughing.	CO1 /CO2
(b)	Explain static, kinetic, and rolling friction with suitable examples.	
Q5.	10 marks	
(a)	Explain how temperature affects viscosity and discuss the concept of the viscosity index.	C01/ CO3
(b)	What are the different types of wear? Describe each with suitable examples.	

